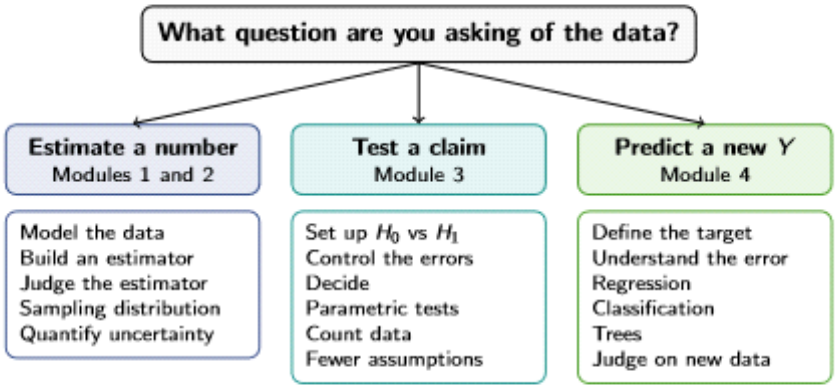
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course_map

The course in one picture



Estimate a number (Modules 1 and 2)

Estimate a number

Model the data: random variables and distributions; a statistical model $\{P_\theta : \theta \in \Theta\}$; assumptions made explicit.

Build an estimator: method of moments; maximum likelihood; Bayesian posterior (posterior mean, MAP).

Judge the estimator: bias; variance and efficiency; MSE = bias^2+ variance; consistency; MLE properties and invariance.

Sampling distribution: CLT; normal, χ^2, t, F; standard error.

Quantify uncertainty: confidence intervals (means, paired data, proportions, variances, variance ratios); credible intervals; bootstrap and jackknife.

Test a claim (Module 3)

Test a claim

- **Set up:** H_0 vs H_1; simple vs composite; one- or two-tailed.
- **Control the errors:** α, β, power; sample size.
- **Decide:** test statistic, its null distribution, rejection region or p-value; duality with confidence intervals.
- **Parametric tests:** z and t for one mean, two means, and paired data; z for proportions; χ^2 for a variance; F for two variances.
- **Count data:** χ^2 goodness of fit, independence, homogeneity.
- **Fewer assumptions:** sign, signed-rank, rank-sum (Mann-Whitney), Kruskal-Wallis, Spearman, permutation tests.

Predict a new Y (Module 4)

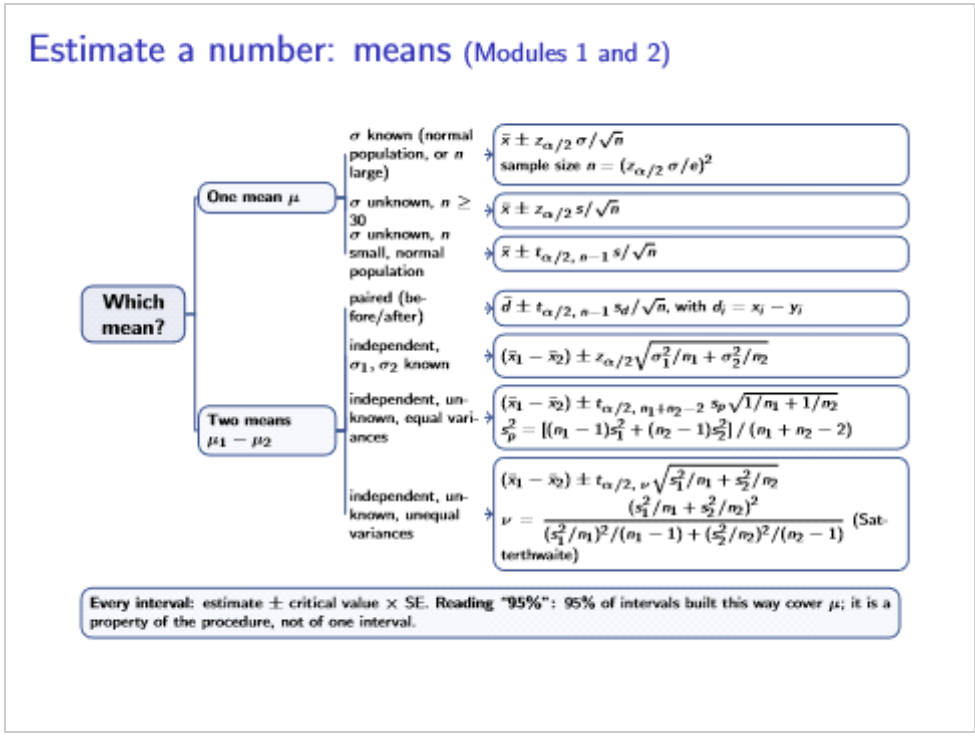
Predict a new Y

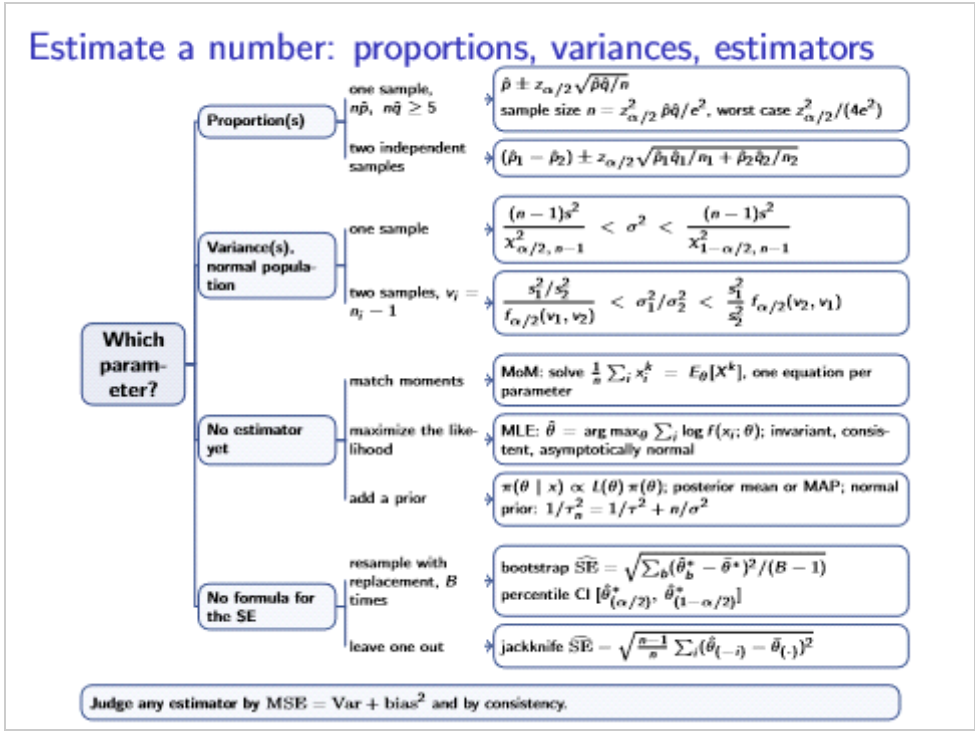
- **Define the target:** loss and risk; the loss picks the answer (mean, median, Bayes classifier).
- **Understand the error:** irreducible + bias^2+ variance; approximation + estimation; the curse of dimensionality.
- **Regression:** least squares as projection; coefficients, dummies, and interactions; ridge and lasso; subset selection.
- **Classification:** logistic regression and GLMs; thresholds, the confusion matrix, ROC and AUC.
- **Trees:** regression and classification trees; pruning; bagging, random forests, boosting.
- **Judge on new data:** test error, validation set, K-fold CV, OOB error.

which_tool_formulas

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which_too
l_formulas





Test a claim: means, proportions, variances (Module 3)

Which parameter?

- **Mean(s)**
  - one sample, σ known → $z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n}) \sim N(0,1)$
  - one sample, σ unknown → $t = (\bar{x} - \mu_0) / (s/\sqrt{n}) \sim t_{n-1}$
  - two independent, σ's known → $z = (\bar{x}_1 - \bar{x}_2 - d_0) / \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2} \sim N(0,1)$
  - two independent, equal variances → $t = (\bar{x}_1 - \bar{x}_2 - d_0) / (s_p\sqrt{1/n_1 + 1/n_2}) \sim t_{n_1+n_2-2}$
  - two independent, unequal variances → $t' = (\bar{x}_1 - \bar{x}_2 - d_0) / \sqrt{s_1^2/n_1 + s_2^2/n_2} \approx t_\nu$ (ν as for the CI)
  - paired → $t = (\bar{d} - d_0) / (s_d/\sqrt{n}) \sim t_{n-1}$
- **Proportion(s)**
  - one sample, small n → $X \sim \mathrm{Bin}(n, p_0)$; p-value from the binomial tail
  - one sample, large n → $z = (\hat{p} - p_0) / \sqrt{p_0 q_0 / n}$
  - two independent samples → $z = (\hat{p}_1 - \hat{p}_2) / \sqrt{\hat{p}\hat{q}\,(1/n_1 + 1/n_2)}$ with $\hat{p} = (x_1 + x_2)/(n_1 + n_2)$
- **Variance(s), normal population**
  - one sample → $\chi^2 = (n-1)s^2/\sigma_0^2 \sim \chi^2_{n-1}$
  - two samples → $f = s_1^2/s_2^2 \sim F_{n_1-1,\, n_2-1}$

Every test: statistic, null distribution, rejection region or p-value; the tail comes from H_1. $\alpha = P(\text{reject } H_0 \mid H_0)$, $\beta = P(\text{keep } H_0 \mid H_1)$, power $= 1 - \beta$; only a larger n lowers both errors. Reject $H_0 : \theta = \theta_0$ iff θ_0 is outside the matching CI.

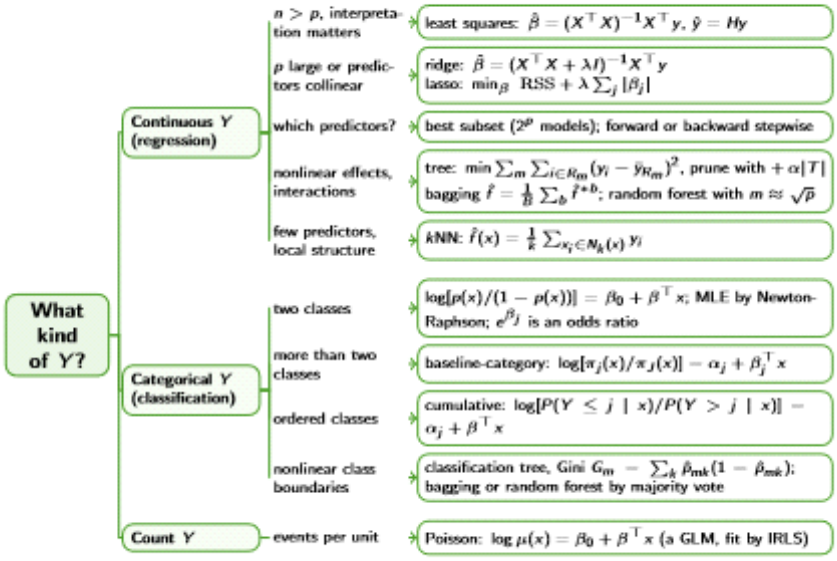
Test a claim: fewer assumptions, count data

What is the claim about?

- **Location, normality doubtful or outliers**
  - one sample or paired: sign test → $X = \#\{+\} \sim \mathrm{Bin}(n, \frac{1}{2})$; large n: $z = (X - n/2) / \sqrt{n/4}$
  - one sample or paired, symmetric: signed-rank → $T = \min(T_+, T_-)$; large n: $z = (T_+ - \mu_T)/\sigma_T$, with $\mu_T = n(n+1)/4$ and $\sigma_T^2 = n(n+1)(2n+1)/24$
  - two independent: rank-sum (Mann-Whitney) → $U_1 = W_1 - n_1(n_1+1)/2$, $U = \min(U_1, U_2)$; large n: $z = (U_1 - n_1 n_2/2) / \sqrt{n_1 n_2 (n_1 + n_2 + 1)/12}$
  - k independent samples → Kruskal-Wallis H: the rank-sum idea for k samples
  - association of x and y → Spearman $r_S = 1 - 6\sum d_i^2 / [n(n^2-1)]$
- **Any statistic**
  - relabel the data under H_0 → $p = \#\{\text{relabelings with } |T^*| \geq |T_{\text{obs}}|\} / \#\text{relabelings}$
- **Counts in categories**
  - one variable vs a stated distribution → $\chi^2 = \sum_i (o_i - e_i)^2/e_i \sim \chi^2_{k-1}$; one df less per estimated parameter; merge small cells
  - two categorical variables: independence or homogeneity → $e_{ij} = (\text{row } i \text{ total})(\text{column } j \text{ total})/n$, $\chi^2 = \sum_{ij}(o_{ij} - e_{ij})^2/e_{ij} \sim \chi^2_{(r-1)(c-1)}$

Nonparametric tests assume less, so they apply more broadly, but they are less efficient when normality holds. Independence is still assumed.

Predict a new Y: choose a model (Module 4)



Predict a new Y: tune and judge on new data



What error?

The loss sets the target: squared loss gives $f^*(x) = E[Y \mid X = x]$, absolute loss the median, 0-1 loss the Bayes classifier.

$E[(Y - \hat{f}(x_0))^2] = \sigma^2 + \mathrm{bias}^2 + \mathrm{Var}$, and σ^2 is irreducible.

Excess risk = approximation + estimation error.

Training error falls as the model grows, so it can never choose the model.

Choosing by cross-validation

Validation set: one split (e.g. 80/20); honest, but wasteful and variable.

K-fold, with $K = 5$ or 10: $\mathrm{CV}_{(K)} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{MSE}_k$ (error rate for classes).

Tunes λ (ridge, lasso), k (kNN), α (tree size), m (random forest), and the subset size.

OOB error (bagging, forests): each point is predicted by the trees that did not use it.

Judging a classifier

The threshold is a separate decision: predict 1 if $\hat{p}(x) > c_{FP}/(c_{FP} + c_{FN})$ (0.5 under 0-1 loss).

Sensitivity $= TP/(TP + FN)$
Specificity $= TN/(TN + FP)$
Precision $= TP/(TP + FP)$

Compare accuracy with the majority-class baseline.

ROC: sensitivity vs 1−specificity over all thresholds; summarize by AUC.

traps

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Traps: estimate a number

- **"Divide by n for the variance."** The $1/n$ estimator has bias $-\sigma^2/n$; S^2 with $n-1$ is unbiased. (HW2 Q6)
- **"Unbiased is always better."** Compare MSE: bias 1 with variance 1 beats bias 0 with variance 4. (P1 Q3 to Q7)
- **"This 95% interval contains μ with probability 0.95."** The procedure covers μ in 95% of samples; a computed interval either does or does not. (HW4 Q4)
- **"A narrower interval means a larger sample."** The width also depends on $\hat{p}$ and on $1-\alpha$. (HW2 Q10, HW4 Q6)
- **"46% means 44% to 48%."** A margin is $z_{\alpha/2}\times$SE, computed from the data, not guessed. (HW2 Q9)
- **"Set $\partial \log L/\partial\theta = 0$."** For Uniform$(0,\theta)$ the likelihood is maximized at the boundary, $\hat{\theta} = X_{(n)}$. (P1 Q19 to Q23)
- **"A larger B makes the bootstrap interval more accurate."** B reduces only Monte Carlo error; the error from a small n remains. (P2 Q13)
- **"Credible and confidence intervals say the same thing."** A credible interval is a probability about θ; a confidence interval is a property of the procedure. (P2 Q14)

Traps: test a claim

- **"H_0 is whatever we want to show."** Work back from the error: a type I error rejects a true H_0. (HW3 Q1, Q8; P2 Q1)
- **"Power is how often we reject H_0."** Power is how often we reject a *false* H_0. Power 0.82 means 18% type II errors and $p < 0.05$ in 82% of samples. (HW4 Q2, Q5)
- **"Intervals say nothing about p."** If 0 is inside the 95% CI but outside the 90% CI, then $0.05 < p < 0.10$. (HW4 Q3)
- **"Before/after data are two samples."** Use the paired test on the differences. (P2 Q6)
- **"Pooled t by default."** Check the variances first (F test); otherwise use Welch. (P2 Q9, Q18)
- **"Use the same SE for the test and the CI of $p_1 - p_2$."** Pooled $\hat{p}$ in the test (it holds under H_0); unpooled in the interval. (P2 Q15, Q17)
- **"Goodness of fit always loses a df."** Subtract only for parameters estimated from the data; a fully specified 9:3:3:1 keeps $k-1$. (P2 Q11)
- **"A significant χ^2 shows that discounts cause purchases."** It shows association, not causation. (P2 Q20)
- **"Skewed data rule out the z test."** Large n rescues it through the CLT; with n near 10 it does not. (P2 Q19)

Traps: predict a new Y

- **"Furnishing adds Rs. 4,500."** With an interaction, the dummy coefficient is the gap at area $= 0$ only. (HW5 Q1)
- **"The fit whose sign flipped is wrong."** Marginal and partial coefficients answer different questions. (HW5 Q3)
- **"One slope or one effect for everyone."** Without an interaction the slope averages over groups; with one, a single effect does not exist. (HW5 Q4, Q6)
- **"Three levels, three dummies."** They sum to the intercept column, so $X^\top X$ is singular. Use $k - 1$. (HW5 Q5)
- **"Code minor, moderate, severe as 1, 2, 3."** This assumes equally spaced levels and one linear effect. (HW5 Q7)
- **"Scaling never matters."** The OLS fit is unchanged; the ridge penalty and gradient descent are not. (HW5 Q8, Q9)
- **"1-NN has zero training error, so it is great."** That shows nothing. $k = n$ predicts $\bar{y}$ everywhere: high bias, low variance. (HW5 Q10)
- **"Our error on the training data is 2%."** The test step is missing, and the number is too optimistic. (HW5 Q11)
- **"The fit passes through every point."** With $p > n$, $\hat{\beta}$ is not unique and the model has overfit. (HW5 Q12)
- **"Standardize, then cross-validate."** Preprocessing must be inside each fold; otherwise the CV error is too small. (HW5 Q14)
- **"2-fold and 10-fold CV agree."** 2-fold trains on half the data, so it reports a larger error. (HW5 Q15)
- **"Accuracy 0.86 is good."** The majority-class baseline was 0.83; look at sensitivity. (L31 Default data)

takeaways

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takeaways

What this subject is about

Learning from data when the data are noisy, incomplete, and finite.

Inference: understand

- Target: a parameter θ of the mechanism that made the data.
- Questions: how large is the effect, how sure are we, which explanation do the data support?
- Judged by: honest uncertainty (bias, SE, coverage, error rates).
- A wrong model gives a wrong conclusion.

Learning: predict

- Target: a function f that maps inputs to a good guess for a new Y.
- Questions: how well will this work on data we have not seen?
- Judged by: error on new data, under the loss that matters.
- A wrong model can still predict well.

Both follow one loop: **model** → **fit** → **quantify uncertainty** → **check** → **decide**.

Think like a statistician

Before any formula

- Ask how the data were collected. The design limits what any analysis can conclude.
- Ask what is missing: who was not sampled, who dropped out, what was not recorded.
- Plot the data first. Summaries can hide everything (Anscombe).
- Variability is the normal state. A number without its uncertainty is incomplete.
- Small samples give extreme results; extremes are followed by less extreme ones.

Before any conclusion

- Correlation is not causation. Randomization, or a design that rules out confounders, is what licenses a causal claim.
- State the assumptions, then check them.
- Ask whether chance alone could produce the pattern.
- Ask "compared with what?" A rate, effect, or accuracy needs a baseline.
- All models are wrong. Ask whether this one is useful for this purpose.

Inference: what to carry away

Estimation

- The sampling distribution is the central object: how would the estimate vary over repeated samples?
- Judge estimators by bias, variance, and $\text{MSE} = \text{bias}^2 + \text{Var}$. A little bias can buy a lot of variance.
- The SE shrinks like $1/\sqrt{n}$: four times the data for half the uncertainty.
- Maximum likelihood is the default workhorse: general, consistent, and efficient in large samples.
- Bayesian inference: prior $\times$ likelihood $\propto$ posterior. With much data, the prior matters little.
- When formulas fail, resample (bootstrap) or simulate.

Intervals and tests

- A 95% CI comes from a procedure that covers the truth 95% of the time. A credible interval is a probability statement about θ.
- A p-value measures surprise under H_0. It is not $P(H_0)$, and it is not the size of the effect.
- Statistical significance is not practical importance. Report the effect size with its interval.
- Not rejecting H_0 is not evidence that H_0 is true. Low power hides real effects.
- Many tests produce false positives by chance. Correct for multiplicity, or fix the analysis before seeing the data.
- Plan the sample size for the power you need, before collecting.

Learning: what to carry away

Generalization

- Expected error = irreducible noise + bias^2 + variance. No method beats the noise floor.
- Flexible models lower bias and raise variance. Training error always falls; error on new data does not.
- Never judge a model on the data used to fit or tune it. Use a held-out test set, touched once, and CV for choices.
- Leakage (anything from the test data entering training, including preprocessing) makes results look better than they are.
- With many predictors and few observations, assume structure: sparsity, smoothness, or regularization.

Building and judging models

- Start with a simple baseline. A complex model must beat it to earn its place.
- Choose the loss and the metric from the decision. The costs of errors set the threshold, not 0.5 by default.
- Accuracy misleads when classes are imbalanced. Look at the confusion matrix, ROC/AUC, and calibrated probabilities.
- Averaging many models (bagging, forests, boosting) usually beats one model.
- A model is valid only for data like its training data. Watch for shift over time and population.
- Prediction is not explanation: a predictive feature need not be a cause.

Practice and communication

Doing the work

- Make it reproducible: scripts, not clicks; fixed seeds; raw data kept untouched.
- Decide the analysis before looking at the outcome when you can. Label exploratory results as exploratory.
- Try to break your own result: a baseline, a sensitivity check, a different method.
- Report what you tried and dropped, not only what worked.

Telling others

- Lead with the answer, then the uncertainty, then the assumptions.
- Give effect sizes in units people understand.
- Say what the data can and cannot support.
- Data describe people: think about privacy, fairness, and who bears the cost of the model's errors.

Ten questions to ask of any claim

1. Where did the data come from, and who is missing?
2. Compared with what?
3. How large is the effect, and how uncertain is it?
4. Could chance alone explain it?
5. Could something else explain it (confounding, selection)?
6. How many comparisons were made before this one was reported?
7. Were the assumptions checked?
8. Was the model judged on data it had not seen?
9. Would the result hold for a different population or time?
10. What decision does it change, and what does a wrong decision cost?

closing

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Where to go next

Topics that build on this course

- **Causal inference:** experiments, confounding, potential outcomes.
- **Bayesian computation:** MCMC and hierarchical models.
- **GLMs and mixed models:** grouped and repeated measurements.
- **Dependent data:** time series and spatial data.
- **High-dimensional statistics:** sparsity and false discovery rates.
- **Unsupervised learning:** PCA, clustering, and density estimation.
- **Deep learning,** seen as flexible function classes fit by empirical risk minimization.
- **Design of experiments:** collecting the right data in the first place.

Books

- Casella and Berger, *Statistical Inference**: the theory, in depth.
- James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning*: learning, with code.
- Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning**: learning, with the mathematics.
- Efron and Hastie, *Computer Age Statistical Inference**: the bridge between the two.
- Agresti and Kateri, *Foundations of Statistics for Data Scientists**: applied, in R and Python.
- Gelman et al., *Bayesian Data Analysis*: Bayesian practice.
- Hernán and Robins, *Causal Inference: What If*: causality.
- Spiegelhalter, *The Art of Statistics*: statistical thinking for everyone.

* on the course syllabus

Thank you

Questions?

"All models are wrong, but some are useful." (George E. P. Box)

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